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Astronomy & Physics Weekly News

Dept. of Applied Physics & Astronomy - University of Sharjah
Compiled by Dr. Ilias Fernini

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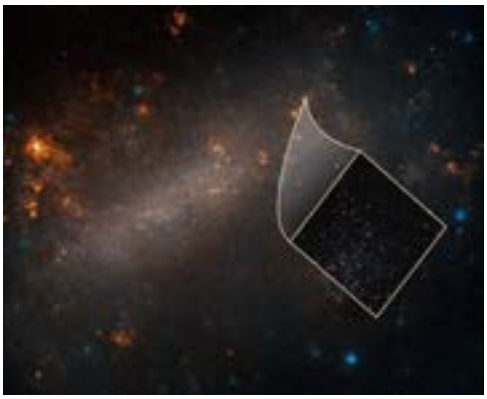
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IT Support Issam Jami

Design by Mohamed Bakir

Mystery of the universe's expansion rate widens with new Hubble data



Ground-based telescope's view of the Large Magellanic Cloud, a satellite galaxy of our Milky Way. The inset image, taken by the Hubble Space Telescope, reveals one of many star clusters scattered throughout the dwarf galaxy. Credit: NASA, ESA, Adam Riess, and Palomar Digitized Sky Survey

Astronomers using NASA's Hubble Space Telescope say they have crossed an important threshold in revealing a discrepancy between the two key techniques for measuring the universe's expansion rate. The recent study strengthens the case that new theories may be needed to explain the forces that have shaped the cosmos.

A brief recap: The universe is getting bigger every second. The space between galaxies is stretching, like dough rising in the oven. But how fast is the universe expanding? As Hubble and other telescopes seek to answer this question, they have run into an intriguing difference between what scientists predict and what they observe.

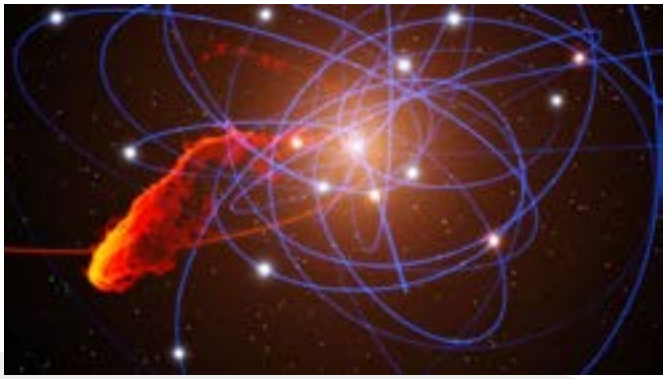
Hubble measurements suggest a faster expansion rate in the modern universe than expected, based on how the universe appeared more than 13 billion years ago. These measurements of the early universe come from the European Space Agency's Planck satellite. This discrepancy has been identified in scientific papers over the last several years, but it has been unclear whether differences in measurement techniques are to blame, or whether the difference could result from unlucky measurements.

The latest Hubble data lower the possibility that the discrepancy is only a fluke to 1 in 100,000. This is a significant gain from an earlier estimate, less than a year ago, of a chance of 1 in 3,000.

These most precise Hubble measurements to date bolster the idea that new physics may be needed to explain the mismatch.

"The Hubble tension between the early and late universe may be the most exciting development in cosmology in decades," said lead researcher and Nobel laureate Adam Riess of the Space Telescope Science Institute (STScI) and Johns Hopkins University, in Baltimore, Maryland. "This mismatch has been growing and has now reached a point that is really impossible to dismiss as a fluke. This disparity could not plausibly occur just by chance." ....Read More...

Physicists Closer to Solving Mystery of Weird Glowing Ring Around Milky Way's Black Hole



This illustration shows the gas passing the black hole. The orange cloud is G2, and the blue lines represent the orbits of nearby stars around the supermassive black hole. The black hole's gravity shapes the behavior of every object in its neighborhood. (Image: © ESO/MPE/Marc Schartmann)

DENVER — Astronomers watched a high-speed gas cloud slam into the matter getting sucked toward Sagittarius A* — the supermassive black hole at the center of the Milky Way — and then zip away into space. Now, careful observations have revealed just how much the gas cloud, which astronomers named G2, slowed after the collision.

That measurement tells scientists something important: the density of the hot matter surrounding Sagittarius A*, which is the nearest known supermassive black hole to Earth. SagittariusA* (SagA*) is quiescent, meaning it's not gobbling up a huge disk of matter and firing off jets. But there's still something hot and glowing surrounding it that physicists don't understand very well. The collision with G2 is offering astronomers one of their best clues yet as to what that glowing ring is made of.

"There was this drag force. The thing [G2] became slower," said Stefan Gillessen, an astronomer at the Max Planck Institute for Extraterrestrial Physics in Garching, Germany.

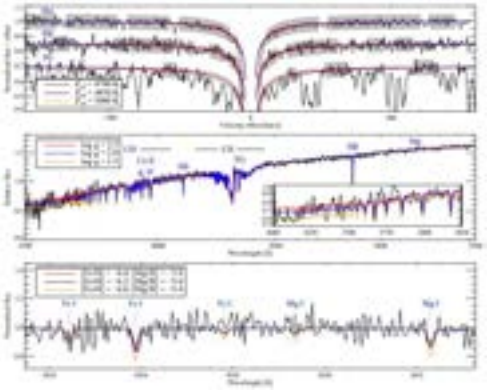
G2's deceleration proved that there was something substantial in the immediate vicinity of the black hole for G2 to crash through, Gillessen said.

Physicists detected that slowing-down using data from the GRAVITY collaboration at the Very Large Telescope (VLT) in Chile. GRAVITY brings together infrared light from all four telescopes of the VLT to make one extra-sharp image. That allowed the researchers an unprecedented view of an object's near-miss with a black hole.

"So of course it was fun to watch, but now we've turned it into something useful," Gillessen told Live Science. "We have actually measured the atmosphere around a black hole at a radius, which was completely inaccessible before."

G2 is a strange object itself: a blobby mass of warm gas that might have a star system or two at its ..Read More...

New ultra metal-poor star discovered



Upper panel: Fit of the effective temperature to the first three Balmer lines (labeled) in the MIKE high-resolution spectrum, compared to models at the preferred Teff = 4850 K. The lines are shown on a velocity scale centred on each line, and have been offset vertically. The grey shaded blocks represent the wavelength ranges used in the χ2 minimisation. Middle panel: Fit of the surface gravity to the WiFeS medium-resolution spectrophotometry, with a zoomed inset showing the Balmer jump region, at the preferred log g = 2.0. Lower panel: Example fits to lines of Fe and Mg in the MIKE high-resolution spectrum. In all panels additional models illustrate the sensitivity, and the legend lists the models as shown from top to bottom. Credit: Nordlander et al., 2019.

An international team of astronomers has detected a new ultra metal-poor star with the lowest ever measured abundance of iron. Designated SMSS J160540.18−144323.1, the newly found object is the most iron-deficient star for which iron has been detected. The finding is presented in a paper published April 16 on the arXiv pre-print repository.

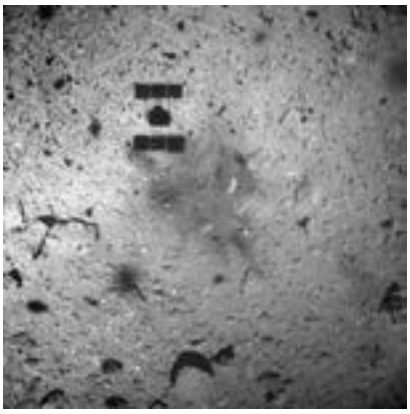
Metal-poor stars are rare objects as only few stars with iron abundances [Fe/H] below -5 have been discovered so far. Currently, SMSS J0313-6708, with metallicity below -7.3, is the most iron-poor star known to date. However, the most iron-poor star where iron has actually been detected is HE 1327−2326 with metallicity at a level of −5.7.

Astronomers are interested in expanding the still short list of metal-poor stars as such objects have the potential to improve our knowledge of the chemical evolution of the universe. The early evolution of the universe is believed to be dependent on the properties of the first generation of metal-free stars.

Searching for such stars using the SkyMapper telescope in Australia and spectrographs on other ground-based telescopes, a group of astronomers led by Thomas Nordlander of Australian National University, has identified a new addition to the list of stars with extremely low metallicities. The observations found that the star SMSS J160540.18−144323.1 (SMSS 1605−1443 for short), located in the Milky Way's halo, most likely some 36,000 light years away, has an ultra low abundance of iron.

"We report the discovery of SMSS J160540.18−144323.1, a new ultra-metal poor halo star discovered with the SkyMapper telescope," the researchers wrote ...Read More...

Japan creates first artificial crater on asteroid



Japan's Hayabusa2 mission aims to shed light on how the solar system evolved

Japanese scientists have succeeded in creating what they called the first-ever artificial crater on an asteroid, a step towards shedding light on how the solar system evolved, the country's space agency said Thursday.

The announcement comes after the Hayabusa2 probe fired an explosive device at the Ryugu asteroid early this month to blast a crater in the surface and scoop up material, aiming to reveal more about the origins of life on Earth.

Yuichi Tsuda, Hayabusa2 project manager at the Japanese space agency (JAXA), told reporters they confirmed the crater from images captured by the probe located 1,700 metres (5,500 feet) from the asteroid's surface.

"Creating an artificial crater with an impactor and observing it in detail afterwards is a world-first attempt," Tsuda said.

"This is a big success."

NASA's Deep Impact probe succeeded in creating an artificial crater on a comet in 2005, but only for observation purposes.

Masahiko Arakawa, a Kobe University professor involved in the project, said it was "the best day of his life".

"We can see such a big hole a lot more clearly than expected," he said, adding the images showed a crater 10 metres in diameter.

JAXA scientists had previously predicted that the crater could be as large as 10 metres in diameter if the surface was sandy, or three metres if rocky.

"The surface is filled with boulders but yet we created a crater this big. This could mean there's a scientific mechanism we don't know or something special about Ryugu's materials," the professor said.

The aim of blasting the crater on Ryugu is to throw up "fresh" material from under the asteroid's surface that could shed light on the early stages of the ....Read More...

Research on disk galaxies sheds light on movement of stars



Credit: University of Arkansas

University of Arkansas astrophysicists have taken an important step toward solving the mystery of how disk galaxies maintain the shape of their spiral arms. Their findings support the theory that these arms are created by a wave of denser matter that creates the spiral pattern as it travels across the galaxy.

"The structure of spiral arms in disk galaxies is a mystery," said Ryan Miller, visiting assistant professor of physics. "No one knows what determines the shape of these spirals, or why they have certain numbers of arms. Our research provides a clear answer to part of that mystery."

Disk galaxies, including the Milky Way, comprise 70 percent of known galaxies. They are characterized by their spiral-shaped arms, but astronomers are not sure how these form and maintain themselves.

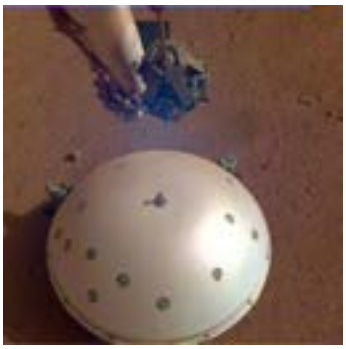
The mystery begins with a simple paradox: stars in a disk galaxy orbit a central mass called a "galactic bulge," and the stars closer to the center orbit faster than the stars toward the edge. But, if the spiral arms were composed of a fixed group of stars, the ones on the edges of the pattern would have to cover more distance than the stars in the middle in order to maintain the spiral pattern. Like runners in the outer lane of a circular track, they would need to move faster to keep their position in the group.

In the 1960s, astronomers proposed the "density wave theory" to explain this paradox. The theory holds that the arms of disk galaxies are not formed from static bundles of stars. Instead, these arms are waves of denser areas that move through the stars. The stars move in accordance to the laws of physics, and as they orbit the center of the galaxy, they encounter these denser areas.

Many astronomers have compared the wave of denser matter to a traffic jam in which the speed of stars traveling in a circle around the center of a galaxy is affected by the denser matter in the same way that motor vehicles are affected by a congested part of a highway. They slow as they meet the congestion and then move more easily after getting past the traffic jam.

The denser areas also affect clouds of gas that pass through these regions.They become ..Read More...

'Marsquake': first tremor detected on Red Planet



This image of InSight's seismometer was taken on the 110th Martian day, or sol, of the mission. The seismometer is called Seismic Experiment for Interior Structure, or SEIS. Credit: NASA/JPL-Caltech

Scientists said Tuesday they might have detected the first known seismic tremor on Mars in a discovery that could shed light on the ancient origins of Earth's neighbour.

A dome-shaped probe known as SEIS landed on the surface of Mars in December after hitching a ride on NASA's InSight spacecraft.

Its instruments measure surface vibrations caused by weather but are also capable of detecting movement from deep within the planet—so called "marsquakes"—or those caused by meteorite impacts.

The French space agency Cnes, which operates SEIS, said it had detected "a weak but distinct seismic signal" from the probe.

The team hopes to be able to gather information about the activity at the centre of Mars, hopefully providing insight into its formation billions of years ago.

"It's great to finally have a sign that there's still seismic activity on Mars," said Philippe Lognonne, a researcher at Paris' Institut de Physique du Globe.

"We've waited for our first Martian quake for months."

According to NASA's Bruce Banerdt, the quake detection "marks the birth of a new discipline: Martian seismology."

The team said they were still working to confirm the cause of the tremor, picked up on April 6, and ensure it came from the planet's interior rather than wind or noise distortion.

It said three other similar but weaker signals of tremors had been picked up by the apparatus. ..Read More...

Rapid destruction of Earth-like atmospheres by young stars



Credit: CC0 Public Domain

The discoveries of thousands of planets orbiting stars outside our solar system has made questions about the potential for life to form on these planets fundamentally important in modern science. Fundamentally important for the habitability of a planet is whether or not it can hold onto an atmosphere, which requires that the atmosphere is not completely lost early in the lifetime of the planet. A new study by researchers based at the University of Vienna and at the Space Research Institute of the ÖAW in Graz has shown that young stars can rapidly destroy the atmospheres of potentially-habitable Earth-like planets, which is a significant additional difficulty for the formation of life outside our solar system. The results will appear soon in the journal Astronomy & Astrophysics Letters.

One of the most active and exciting questions in modern science is how abundant planets with Earth-like atmospheres and surface conditions and therefore the potential for harbouring life are in the universe. Much recent research on this topic has focused on planets orbiting stars called M-dwarfs, which are smaller than our Sun and are the most numerous type of star in our solar neighborhood.

The primary driver of atmospheric losses to space is the central star that the planet is orbiting. Stars have strong magnetic fields, and these lead to the emission of high energy X-ray and ultraviolet radiation. These phenomena are known collectively as the star's 'activity.' At young ages, stars have high levels of activity, and therefore emit extremely large amounts of X-rays and ultraviolet radiation. As stars age, their activities decrease rapidly. Importantly for planets orbiting M-dwarfs, while the activities of stars like the Sun decrease rapidly after a few hundred million years, M-dwarfs often remain highly active for billions of years.

The high energy radiation is important because it is absorbed high in the atmosphere of a planet, causing the gas to be heated. For the Earth, the gas is heated to temperatures of more than 1000 degrees Celsius in the upper region known as the thermosphere. This is the region in which spacecraft such as satellites and the International Space Station fly. When orbiting young stars with high activity levels, the thermospheres of planets ..Read More...

The giant galaxy around the giant black hole



The galaxy M87, imaged here by NASA's Spitzer Space Telescope, is home to a supermassive black hole that spews two jets of material out into space at nearly the speed of light. The inset shows a close-up view of the shockwaves created by the two jets. Credit: NASA/JPL-Caltech/IPAC

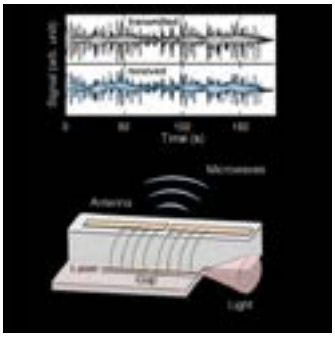
On April 10, 2019, the Event Horizon Telescope (EHT) unveiled the first-ever image of a black hole's event horizon, the area beyond which light cannot escape the immense gravity of the black hole. That giant black hole, with a mass of 6.5 billion Suns, is located in the elliptical galaxy Messier 87 (M87). EHT is an international collaboration whose support in the U.S. includes the National Science Foundation.

This image from NASA's Spitzer Space Telescope shows the entire M87 galaxy in infrared light. The EHT image, by contrast, relied on light in radio wavelengths and showed the black hole's shadow against the backdrop of high-energy material around it.

Located about 55 million light-years from Earth, M87 has been a subject of astronomical study for more than 100 years and has been imaged by many NASA observatories, including the Hubble Space Telescope, the Chandra X-ray Observatory and NuSTAR. In 1918, astronomer Heber Curtis first noticed "a curious straight ray" extending from the galaxy's center. This bright jet of high-energy material, produced by a disk of material spinning rapidly around the black hole, is visible in multiple wavelengths of light, from radio waves through X-rays. When the particles in the jet impact the interstellar medium (the sparse material filling the space between stars in M87), they create a shockwave that radiates in infrared and radio wavelengths of light but not visible light. In the Spitzer image, the shockwave is more prominent than the jet itself.

The brighter jet, located to the right of the galaxy's center, is traveling almost directly toward Earth. Its brightness is amplified due to its high speed in our direction, but even more so because of what scientists call "relativistic effects," which arise because the material in the jet is traveling near the speed of light. The jet's trajectory is just slightly offset from our line of sight with respect to the galaxy, so we can still see some of the length of the jet. The shockwave begins around the point where the jet appears to curve down, highlighting the regions where the fast-moving particles are colliding with ..Read More...

Researchers transmit data via a semiconductor laser, opening the door to ultra-high-speed Wi-Fi



This device uses a frequency comb laser to emit and modulate microwaves wirelessly. The laser uses different frequencies of light beating together to generate microwave radiation. The researchers used this phenomenon to send a song wirelessly to a receiver. Credit: Marco Piccardo/Harvard SEAS

You've never heard Dean Martin like this.

Researchers from the Harvard John A. Paulson School of Engineering and Applied Sciences transmitted a recording of Martin's classic "Volare" wirelessly via a semiconductor laser—the first time a laser has been used as a radio frequency transmitter.

In a paper published in the Proceedings of the National Academy of Sciences, the researchers demonstrated a laser that can emit microwaves wirelessly, modulate them, and receive external radio frequency signals.

"The research opens the door to new types of hybrid electronic-photonic devices and is the first step toward ultra-high-speed Wi-Fi," said Federico Capasso, the Robert L. Wallace Professor of Applied Physics and Vinton Hayes Senior Research Fellow in Electrical Engineering, at SEAS and senior author of the study.

This research builds on previous work from the Capasso Lab. In 2017, the researchers discovered that an infrared frequency comb in a quantum cascade laser could be used to generate terahertz frequencies, the submillimeter wavelengths of the electromagnetic spectrum that could move data hundreds of times faster than today's wireless platforms. In 2018, the team found that quantum cascade laser frequency combs could also act as integrated transmitters or receivers to efficiently encode information.

Now, the researchers have figured out a way to extract and transmit wireless signals from laser frequency combs.

Unlike conventional lasers, which emit a single frequency of light, laser frequency combs emit multiple frequencies simultaneously, evenly spaced to resemble the teeth of a comb. In 2018, the researchers discovered that inside the laser, the different frequencies of light beat together to generate microwave radiation. The light inside the cavity of the laser caused electrons to oscillate at microwave frequencies—which are within ...Read More...

Dark matter detector observes rarest event ever recorded



The XENON1T dark matter collaboration has observed the radioactive decay of xenon-124, which has a half-life of 1.8 X 1022 years

How do you observe a process that takes more than one trillion times longer than the age of the universe? The XENON Collaboration research team did it with an instrument built to find the most elusive particle in the universe - dark matter.

In a paper to be published tomorrow in the journal Nature, researchers announce that they have observed the radioactive decay of xenon-124, which has a half-life of 1.8 sextillion (that's 18 followed by 21 zeros) years.

"We actually saw this decay happen. It's the longest, slowest process that has ever been directly observed, and our dark matter detector was sensitive enough to measure it," said Ethan Brown, an assistant professor of physics at Rensselaer, and co-author of the study. "It's an amazing to have witnessed this process, and it says that our detector can measure the rarest thing ever recorded."

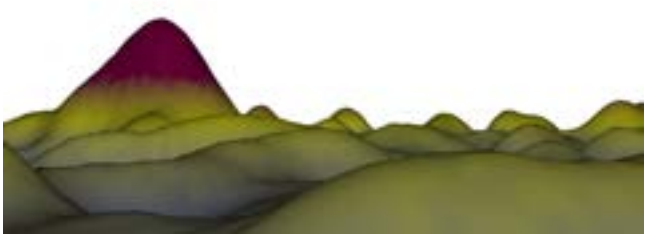
The XENON Collaboration runs XENON1T, a 1,300-kilogram vat of super-pure liquid xenon shielded from cosmic rays in a cryostat submerged in water deep 1,500 meters beneath the Gran Sasso mountains of Italy.

The researchers search for dark matter (which is five times more abundant than ordinary matter, but seldom interacts with ordinary matter) by recording tiny flashes of light created when particles interact with xenon inside the detector. And while XENON1T was built to capture the interaction between a dark matter particle and the nucleus of a xenon atom, the detector actually picks up signals from any interactions with the xenon.

The evidence for xenon decay was produced as a proton inside the nucleus of a xenon atom converted into a neutron. In most elements subject to decay, that happens when one electron is pulled into the nucleus. But two protons in a xenon atom must simultaneously absorb two electrons to convert into two neutrons, an event called "double-electron capture."

Double-electron capture only happens when two of the electrons are right next to the nucleus at just the right time, Brown said, which is "a rare thing multiplied by another rare thing, making it ultra-rare." ...Read More...

Planck reveals link between active galaxies and their dark matter environment



Gravitational deflection by quasar-hosting dark matter halos. Credit: David Tree, Professor Peter Richardson, Games and Visual Effects Research Lab, University of Hertfordshire

Scientists have used the tiny distortions imprinted on the cosmic microwave background by the gravity of matter throughout the universe, recorded by ESA's Planck satellite, to uncover the connection between the luminosity of quasars – the bright cores of active galaxies – and the mass of the much larger 'halos' of dark matter in which they sit. The result is an important confirmation for our understanding of how galaxies evolve across cosmic history.

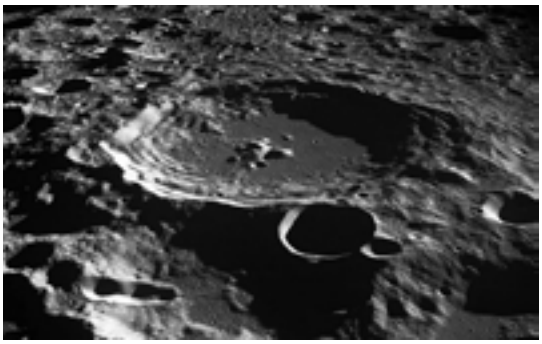
Most galaxies in the universe are known to host supermassive black holes, with masses of millions to billions of times the Sun's mass, at their cores. The majority of these cosmic monsters are 'dormant', with little or no activity going on near them, but about one percent are classified as 'active', accreting matter from their surroundings at very intense rates. This accretion process causes material in the black hole's vicinity to shine brightly across the electromagnetic spectrum, making these active galaxies, or quasars, some of the brightest sources in the cosmos.

While it is still unclear what activates these black holes, switching on and off their phase of intense accretion, it is likely that quasars play an important role in regulating the evolution of galaxies across cosmic history. For this reason, it is crucial to understand the relationship between quasars, their host galaxies, and their environment on even larger scales.

In a recent study led by James Geach of the University of Hertfordshire, UK, scientists have combined data from ESA's Planck mission with the largest survey of quasars available to date to shed light on this fascinating topic.

According to the leading scenario of structure formation in the universe, galaxies take shape out of ordinary matter in the densest knots of the cosmic web – a filamentary network, made up primarily of the invisible dark matter, that pervades the cosmos. In turn, the complex distribution of both ordinary and dark matter originates from tiny fluctuations in the primordial universe, which leave an imprint in the cosmic microwave background (CMB), the most ancient light in the history of the universe. ...Read More...

The Moon's Surface Is Totally Cracked



An oblique view of the Daedalus Crater on the far side of the moon, as seen from the Apollo 11 spacecraft in lunar orbit. (Image: © NASA)

Is the moon all it's cracked up to be? Yes — and then some. New analysis of the lunar surface reveals that it's far more fractured than once thought.

Since the moon formed 4.3 billion years ago, asteroid impacts have scarred its face with pits and craters. But the damage goes far deeper than that, with cracks extending to depths of 12 miles (20 kilometers), researchers recently reported.

Though the moon's craters have been well-documented, scientists previously knew little about the upper region of the moon's crust, the megaregolith, which sustained the bulk of the damage from space rock bombardment. In the new study, computer simulations revealed that impacts from single objects could fragment the lunar crust into blocks about 3 feet (1 meter) wide, opening surface cracks that extend for hundreds of kilometers. This suggests that much of the fracturing in the megaregolith could have come from single, high-speed impacts, leaving the crust "thoroughly fractured" early in the moon's history. [When Space Attacks: 6 Craziest Meteor Impacts]

These findings helped to address questions raised by NASA's Gravity Recovery and Interior Laboratory (GRAIL), a mission that sent twin spacecraft to the moon in 2011 to create the most detailed lunar gravity map to date.

Data gathered by GRAIL showed that the moon's crust was far less dense than expected, Sean Wiggins, lead author of the new study and a doctoral candidate with the Earth, Environmental and Planetary Sciences Department at Brown University in Rhode Island, told Live Science.

Wiggins and his colleagues suspected that ancient impacts could have substantially fractured the lunar surface, "adding porosity and therefore lowering the density," he said.

Deep impacts

Using simulations, the study authors found that an impact from an object measuring just 0.6 miles (1 km) in diameter could have opened cracks reaching depths of 12 miles (20 km) in the lunar surface. ..Read More...

Special Read:

NASA's Twins Study: Spaceflight changes the human body, but only temporarily



Retired astronaut Mark Kelly (left) cracks a slight smile while posing with his identical twin brother, astronaut Scott Kelly (right). As part of NASA's Twins Study, Scott took a long trip to space, while Mark remained on Earth. Researchers then monitored how their bodies reacted to their differing environments. NASA

A year in space can trigger many changes in the human body — from our heads down to our genes — but most of them don't persist indefinitely.

Brothers compete. So in 2016, when astronaut Scott Kelly returned to Earth after spending a year in space, it must have really annoyed his identical twin brother — retired astronaut Mark Kelly — that Scott was two inches taller than when he left. However, Scott's temporary increase in height was not the only thing that changed during his trip.

As part of NASA's Twins Study, while Scott was in space, Mark went about his daily life on Earth. Over the course of the year-long mission, researchers tracked changes in both brothers' biological markers to pinpoint any variances. Because the twins share the same genetic code, researchers reasoned that any observed differences could tentatively — though not definitively — be linked to Scott's time aboard the International Space Station (ISS). This allowed them to take advantage of a unique opportunity and explore how an extended stay in space may impact the human body.

Based on their results, which were published this week in the journal Science, spaceflight can definitely trigger changes in the human body. But the vast majority of these changes disappear within just a few short months of returning to Earth.

Most notably, the researchers found that living in a microgravity environment can: damage DNA; impact the way thousands of individual genes are expressed; increase the length of telomeres (the shielding caps that protect the ends of our chromosomes); thicken artery walls; modify the microbiome; and increase inflammation — just to name a few.

"This is the dawn of human genomics in space," said Andrew Feinberg, a distinguished professor at Johns Hopkins University and one of the lead investigators for the Twins Study, in a press release. "We developed the methods for doing these types of human genomic studies, and we should be doing more research to draw conclusions about what happens to humans in space."

A smattering of studies

The Twins Study is actually composed of 10 distinct projects all wrapped into one. The individual investigations focus on how long-term spaceflight can impact different aspects of the human body, ranging from genetic expression to cardiovascular health to immune system response and more.

To carry out the study, researchers performed numerous tests and collected many samples from both twins, beginning six months before Scott's flight and ending nine months after he landed. While in orbit, Scott was responsible for collecting his own samples — which included blood, urine, and fecal matter.

Spaceflight changes gene expression

One of the most intriguing results from the Twins Study came from investigating how gene expression (or epigenetics) changed over the duration of the mission. Led by Chris Mason of Weill Cornell Medicine, researchers carried out whole-genome sequencing to identify chemical changes that occurred in the twins' DNA and RNA. ...Read More...

This Week's Sky at a Glance - Apr. 27 - May 03, 2019

Apr. 27	Sa	02:18	Last Quarter
Apr. 28	Su	22:20	Moon Apogee: 404600 km
May 02	Th	15:39	Moon-Venus: 3.9° N

First Dish of SAASST 40-m Radio Interferometer Installed

The Sharjah Academy for Astronomy, Space Sciences, and Technology has recently installed the first 5-m SPIDER 500A dish of the future 40-m radio interferometer. The SAASST interferometer has a rectangular scalene triangle shape, with sides of 30, 40 and 50 meters. This array will be able to simulate the resolution of a 40-m diameter single dish antenna, with a collecting area equal to an antenna of 8.7 meters in diameter. The array will mainly work at 1.4 GHz. The synthesized beam in this configuration measures about 0.36° (21 arc minutes).

